School of Future Tech

Case Study Report

on

FitFlex – Gym Class Scheduler Web Application

by

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1. Introduction to the Case Study

The FitFlex Gym Class Scheduler is a front-end web application designed to simulate an online booking system for fitness classes such as Yoga, HIIT, and Pilates. The project demonstrates how a user-friendly interface can allow users to explore available classes, check schedules, and complete bookings efficiently.

This case study focuses on building a responsive and interactive platform using HTML, CSS, and JavaScript. The system allows users to navigate weekly schedules, filter classes based on difficulty levels, check class capacity, and complete a booking process.

The project demonstrates practical web development concepts such as responsive layouts, DOM manipulation, event handling, and user interaction through a modern and visually appealing user interface.

2. Problem Statement / Case Background (Abstract)

Background

Fitness studios and gyms often need an efficient system to manage class schedules and bookings. Traditional manual booking systems can be inefficient and difficult to manage, especially when tracking available spots and handling multiple classes.

A digital booking platform allows users to easily explore available sessions, register for classes, and manage their schedules more efficiently.

Abstract

This case study presents the design and implementation of a Gym Class Scheduler named FitFlex. The system allows users to view a weekly class schedule, filter classes based on difficulty levels, and book available classes.

The application includes features such as class capacity management, pricing options, discount calculations, instructor profiles, and an interactive booking form. The project demonstrates how modern front-end technologies can be used to create an efficient and user-friendly scheduling system.

3. Case Study Design

The FitFlex system is designed as a multi-page web application with interactive components.

The design consists of the following modules:

1. Schedule Page

Displays weekly fitness classes including:

- Yoga
- HIIT
- Pilates

Users can view instructor information, difficulty level, and available spots.

2. Instructor Page

Displays profiles of instructors including:

- Instructor name
- Specialty
- Experience
- Profile image

3. Booking Page

Allows users to register for a selected class by filling out a booking form.

4. Navigation System

A navigation bar allows users to move between the schedule page, instructor page, and booking page.

5. Filtering System

Users can filter classes by difficulty level such as beginner, intermediate, or advanced.

4. Methods & Algorithms Technology Applied in the Problem Statement

The following technologies and techniques were used to build the FitFlex application.

1. HTML5

HTML is used to structure the web pages including the layout for schedules, instructor profiles, and booking forms.

2. CSS3

CSS is used for styling and visual design. The application uses:

- Flexbox layouts
- Card-based UI design
- Hover animations
- Custom checkbox styling

- Dark theme interface

3. JavaScript

JavaScript provides interactivity and dynamic functionality such as:

- Week navigation
- Class capacity tracking
- Difficulty filtering
- Price calculation
- Discount application
- Booking redirection

4. localStorage

localStorage is used to store the selected class name and pass it between pages so the booking form automatically fills the chosen class.

5. Image Integration

Images from Unsplash are used to create realistic fitness-related visuals.

5. Problem Statement / Case Study Implementation Details and Snapshots.

The project is implemented using multiple files for better organization.

Project Structure

```
FitFlex/
 │
 ├── schedule.html
 ├── instructors.html
 ├── class-booking.html
 ├── style.css
 ├── script.js
```

Implementation Modules

Schedule Page
Displays class cards with images, instructor details, difficulty level, and available spots.

Booking System
When a user clicks the “Book Class” button:

1. Available spots decrease by one.

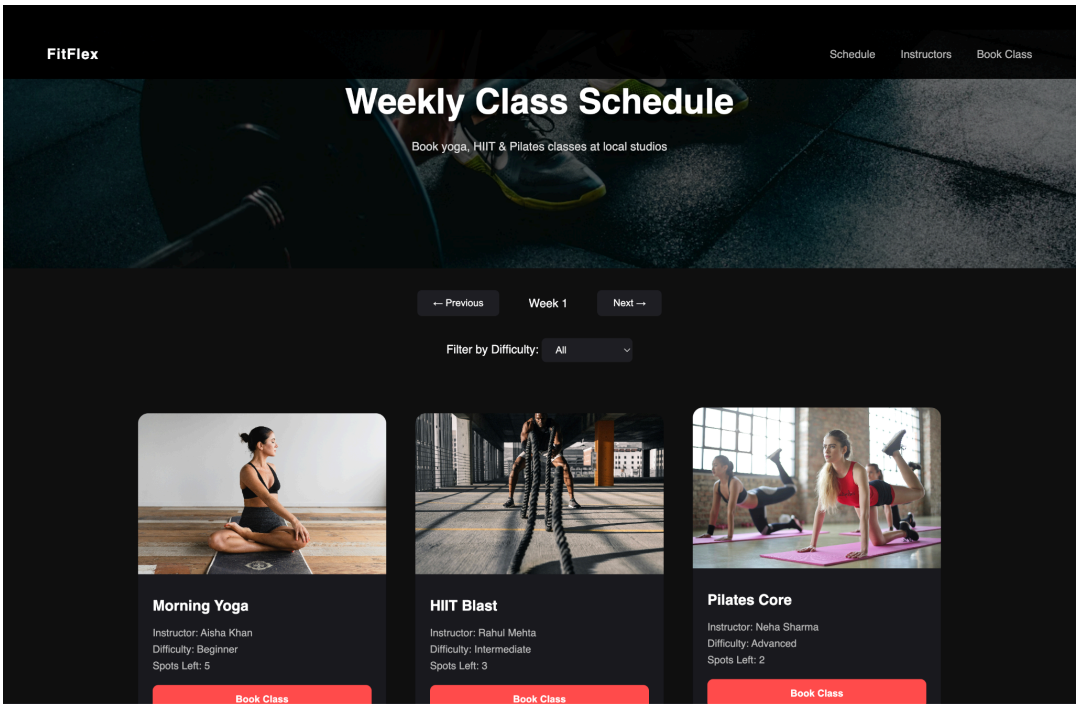
2. The selected class name is stored using localStorage.
3. The user is redirected to the booking page.

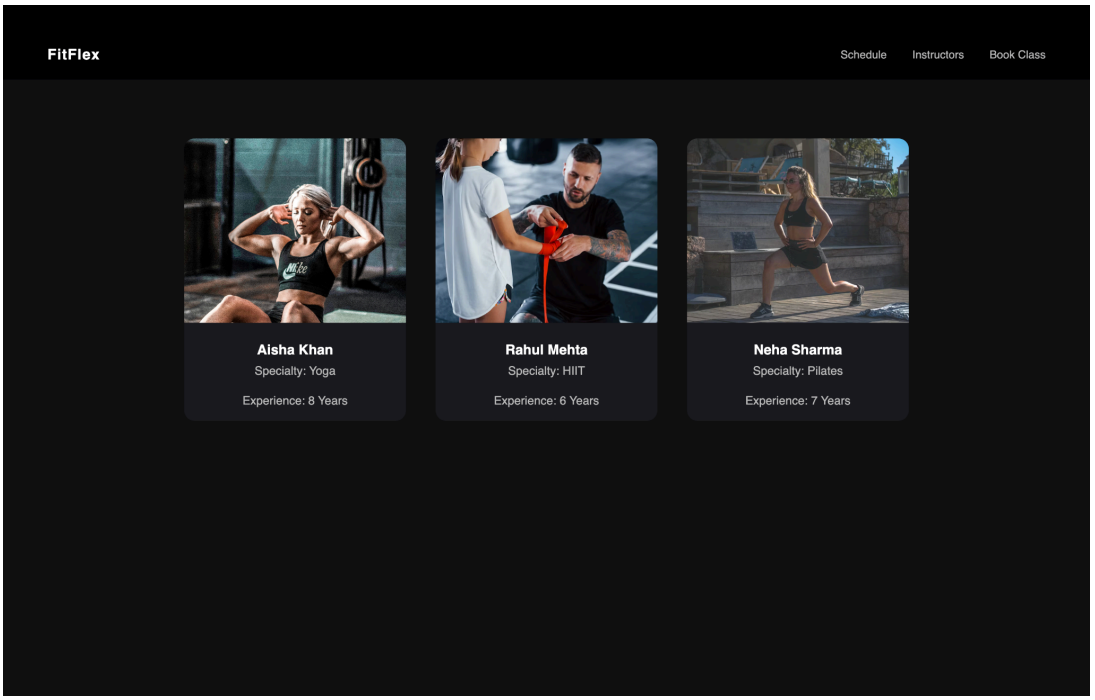
Booking Form

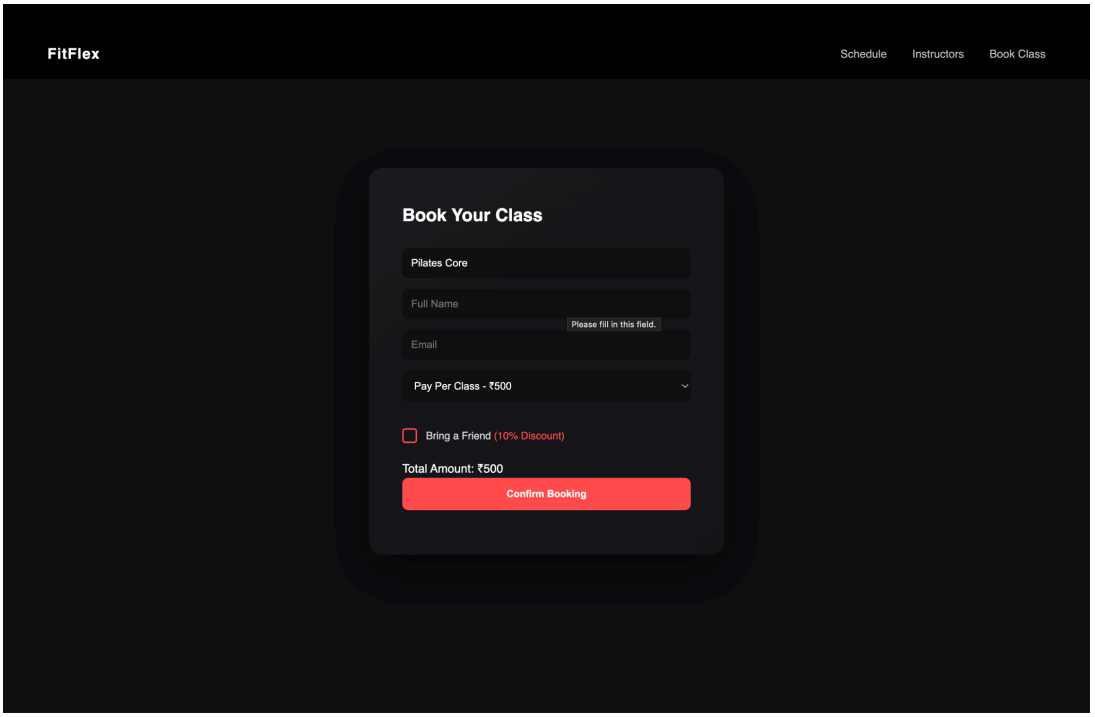
The booking page includes:

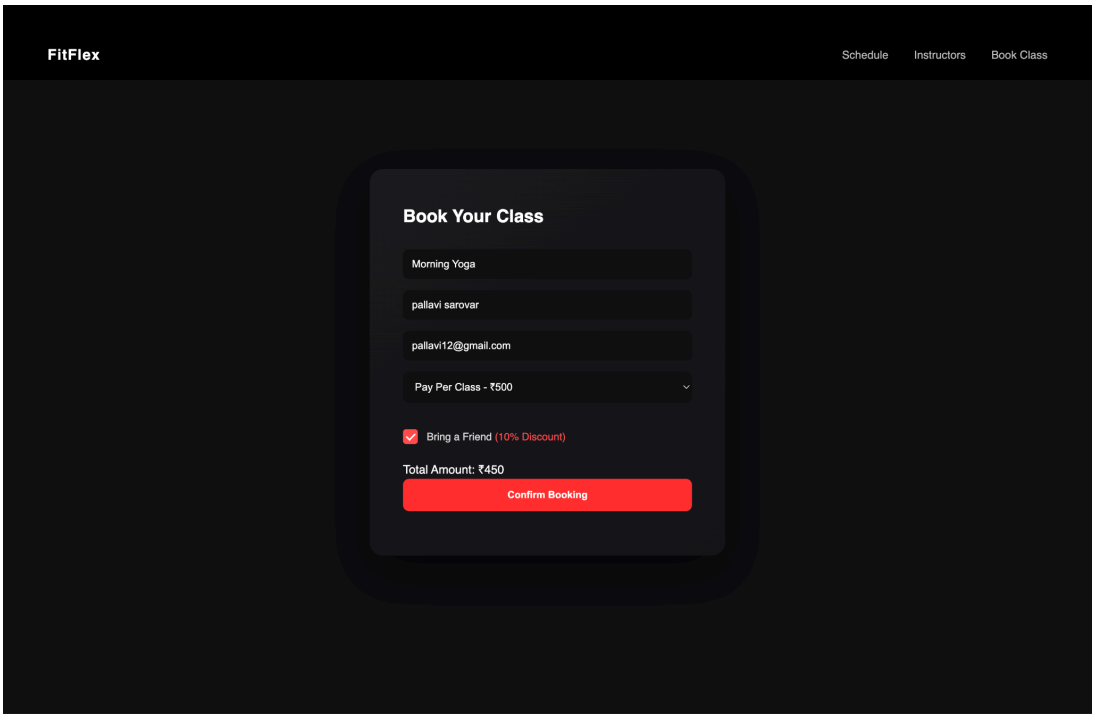
- Selected class field
- Name input
- Email input
- Pricing options
- Discount checkbox
- Total price calculation

Snapshots:




FitFlex
Schedule
Instructors
Book Class
Aisha Khan
Specialty: Yoga
Experience: 8 Years
Rahul Mehta
Specialty: HIIT
Experience: 6 Years
Neha Sharma
Specialty: Pilates
Experience: 7 Years


FitFlex
Schedule
Instructors
Book Class
Book Your Class
Pilates Core
Full Name
Please fill in this field.
Email
Pay Per Class - ₹500
Bring a Friend (10% Discount)
Total Amount: ₹500
Confirm Booking



6. Case Study Results and Conclusion

Results

The FitFlex application successfully demonstrates an interactive gym class booking system. Users can easily explore available classes, filter sessions based on difficulty levels, and register for classes using a simple booking form.

The integration of JavaScript allows real-time interaction including capacity updates, dynamic filtering, and automatic price calculation.

Conclusion

This case study demonstrates how front-end web technologies can be used to develop a practical scheduling and booking system. The project combines responsive design, modern UI practices, and interactive features to create a functional fitness booking platform.

The implementation fulfills the key requirements of the case study including weekly scheduling, difficulty filtering, capacity tracking, discount calculation, instructor profiles, and booking functionality.

7. References

- HTML5 Documentation – https://developer.mozilla.org

- CSS3 Documentation – https://developer.mozilla.org
- JavaScript Documentation – https://developer.mozilla.org
- Unsplash Image Library – https://unsplash.com
- Web Development Tutorials and Learning Resources